Computer Vision – Lecture 2

Image Processing I

10.04.2025

Flipped classroom – Q&A session

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Computer Vision

"Flipped Classroom" Model

- First block of the lecture: Image Processing
  - Pre-recorded videos of the lecture contents are made available before the lecture slots
  - You are expected to watch and rehearse the lecture contents ahead of the lecture (!)
  - The lecture itself will then take the form of a Q&A session
    - Short repetition + opportunity to ask questions

- After that, we will switch back to a regular lecture model
  - Regular lecture format, held in-presence
  - Lecture recordings will be made available as videos

“Flipped Classroom” Model

Lecture 2: Image Processing I (11.04.2022, online)

This lecture slot will be given online. Please use the following permanent zoom link to attend:

https://rwth.zoom.us/j/98615042972?pwd=enNvUlZlZjNNQ2VkMW9zV3E2TC9vUT09

Passcode: 914166

cv22-part02-image-processing1.pdf
2.2MB Hochgeladen 8.04.2022 23:11

cv22-part02-image-processing1-6on1.pdf
1.5MB Hochgeladen 8.04.2022 23:12

Pre-recorded videos

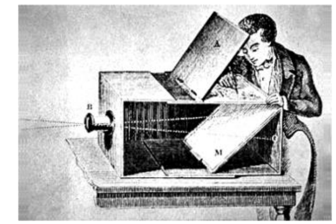
"Flipped classroom" model: Please watch the following pre-recorded videos before the lecture on Monday 11.04. The lecture slot will then contain a Q&A on the topics covered in the videos.

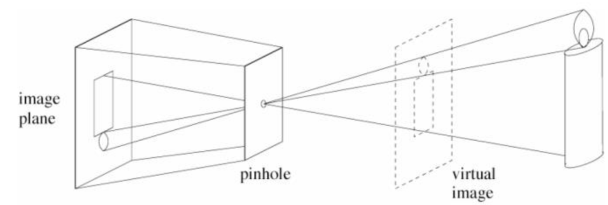
cv22-part02a-image-processing1-qa.pdf
1.3MB Hochgeladen 8.04.2022 23:44

- Pre-recorded videos made available on moodle
  - Part 1.1: Image Formation
  - Part 2.1: Linear Filters
  - Part 2.2: Background on Linear Filters
- *Quick poll: Who watched the videos?*

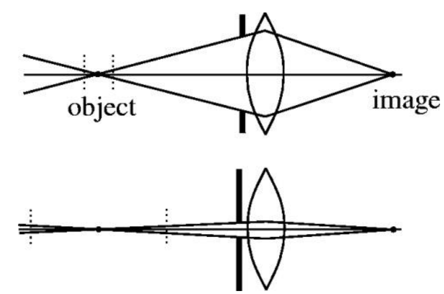
Course Outline

- Image Processing Basics
  - Image Formation
  - Linear Filters
  - Multi-scale Representations
  - Edge & Structure Extraction
- Segmentation & Grouping
- Object Recognition
- Deep Learning Approaches
- 3D Reconstruction

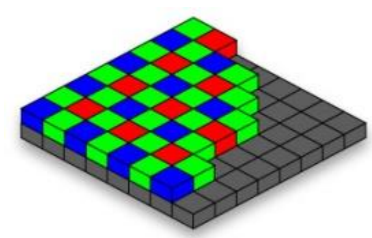




Pinhole camera model



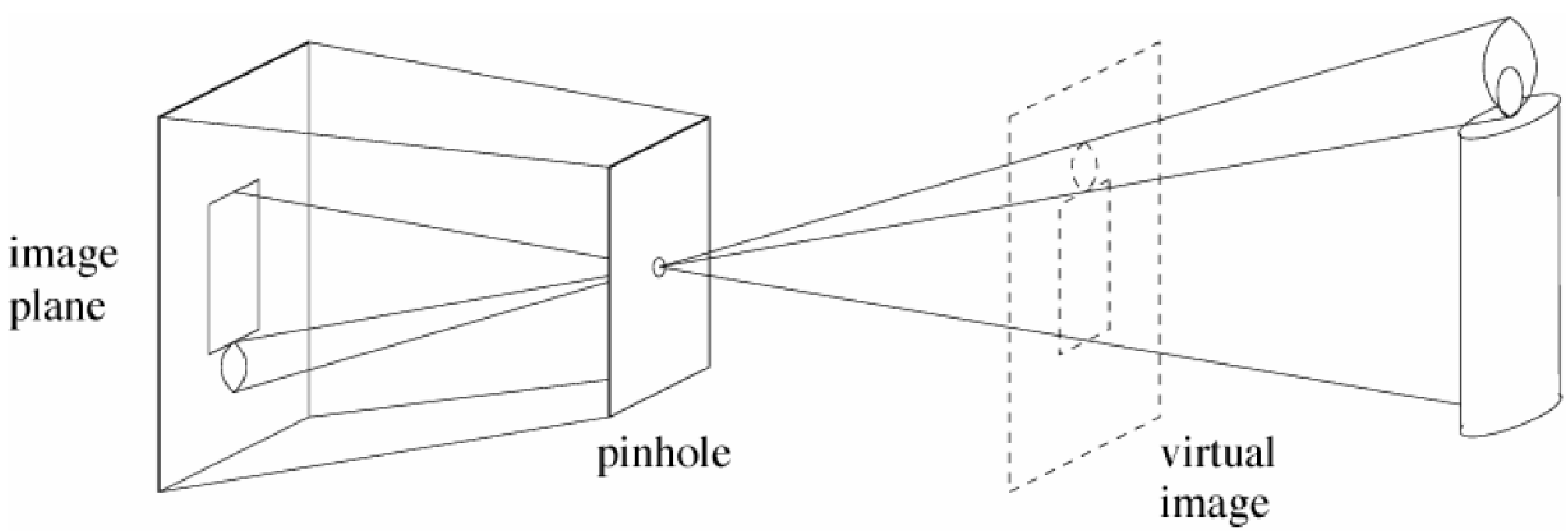
Lenses, focal length, aperture



Color sensors

Recap: Pinhole Camera

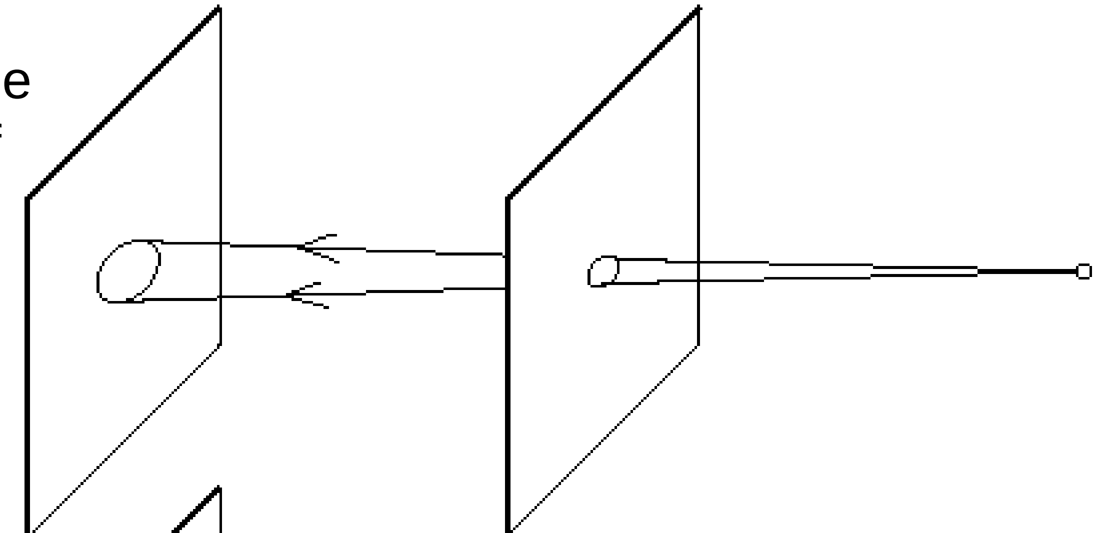
- (Simple) standard and abstract model today
  - Box with a small hole in it
  - Works in practice



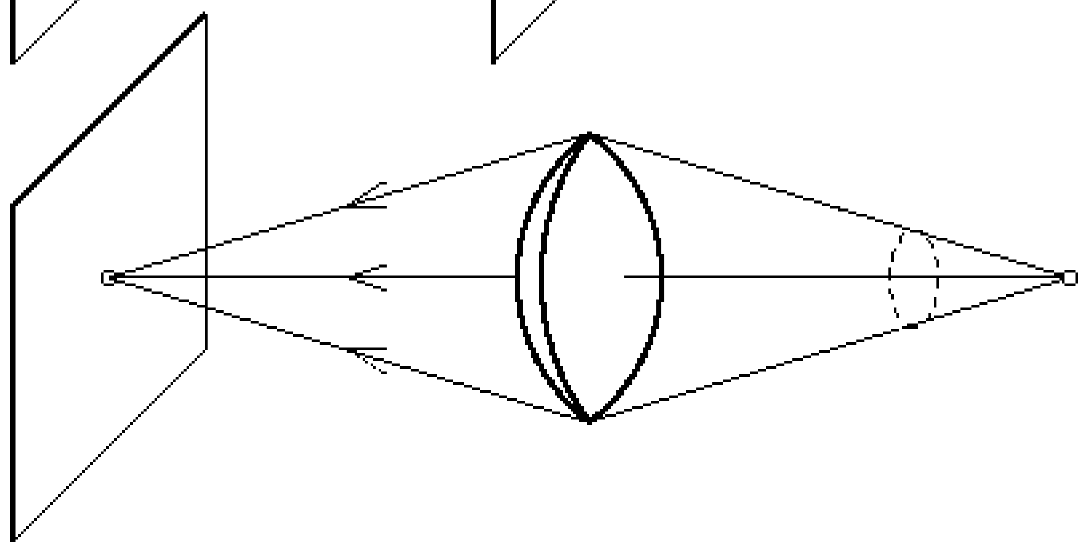
Source: Forsyth & Ponce

Recap: The Reason for Lenses

- Increasing the pinhole size to increase the amount of light causes blur.

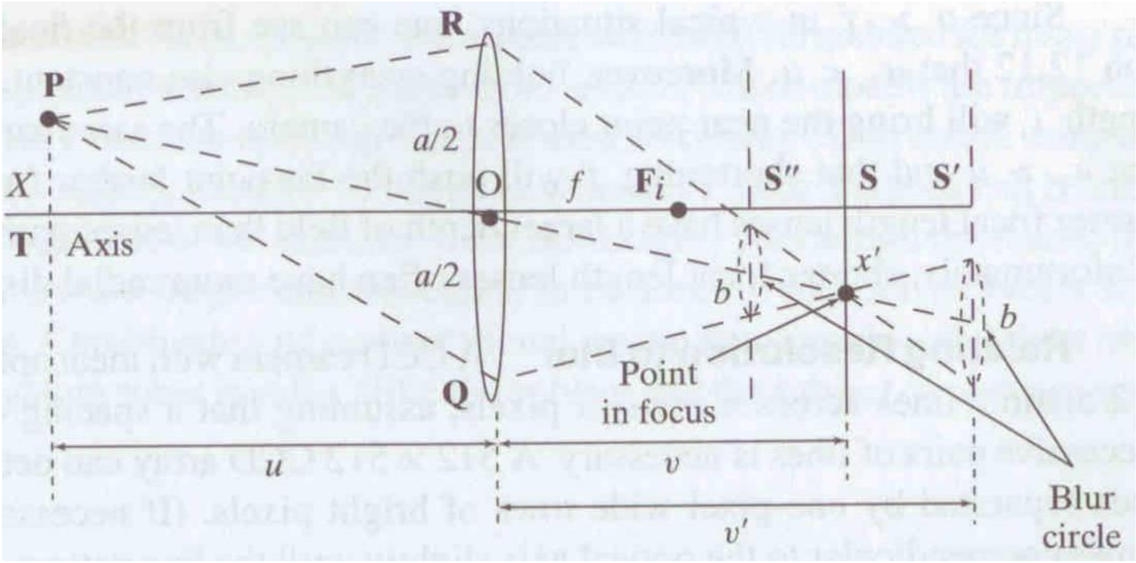


- Lenses keep the image in sharp focus while gathering light from a larger area.



Source: Forsyth & Ponce

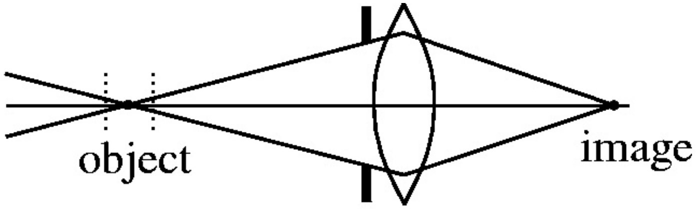
Recap: Focus and Depth of Field

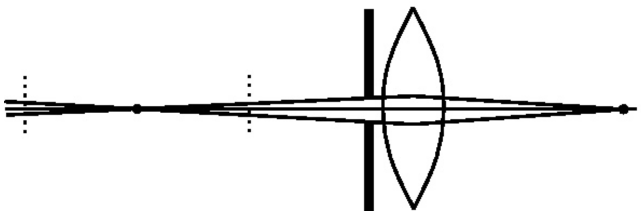


Thin lens: scene points at distinct depths come in focus at different image planes.

This creates blur circles ("circles of confusion")

- Depth of field
  - Distance between image planes where blur is tolerable
  - Affected by aperture



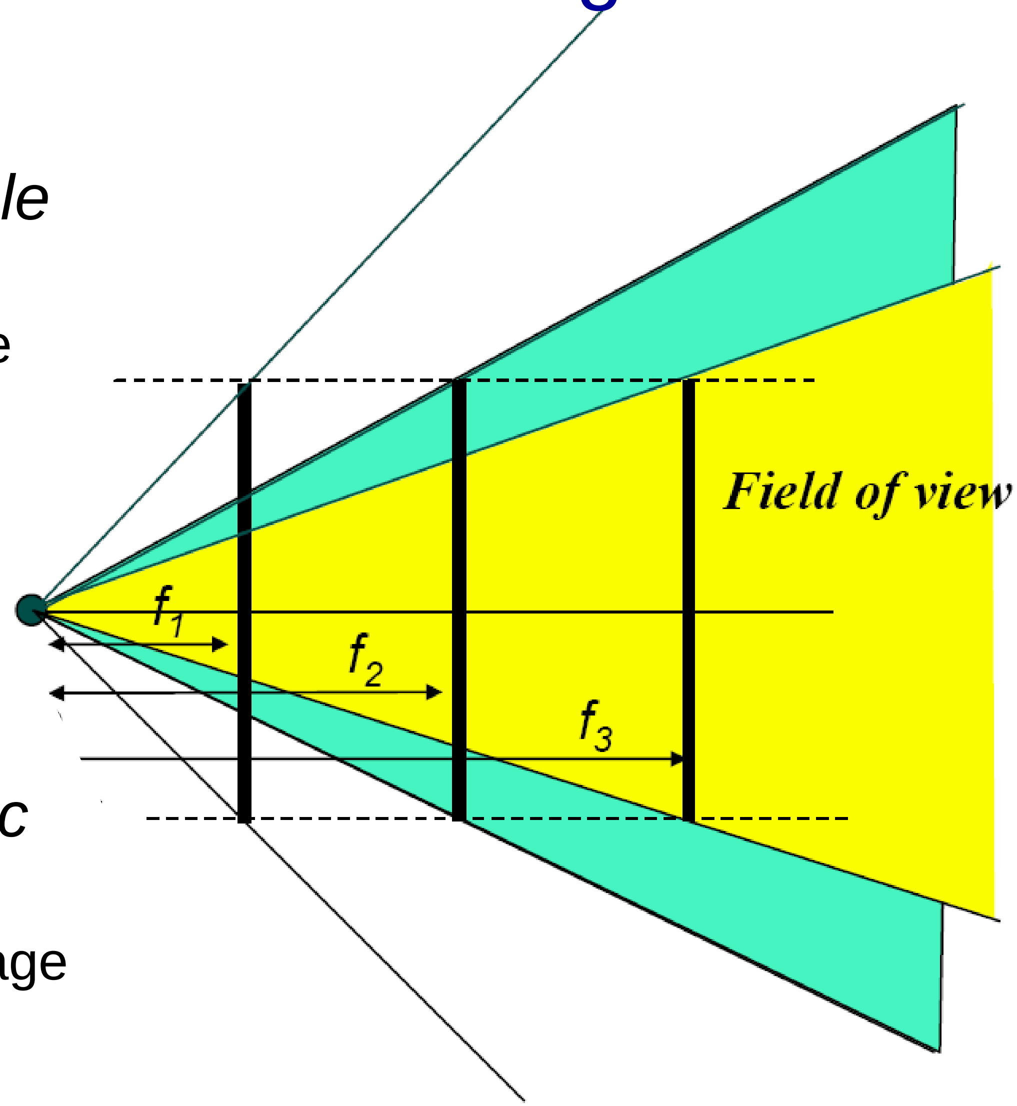


Source: Shapiro & Stockman

Recap: Field of View and Focal Length

- As *f* gets smaller, image becomes more *wide angle*
  - More world points project onto the finite image plane

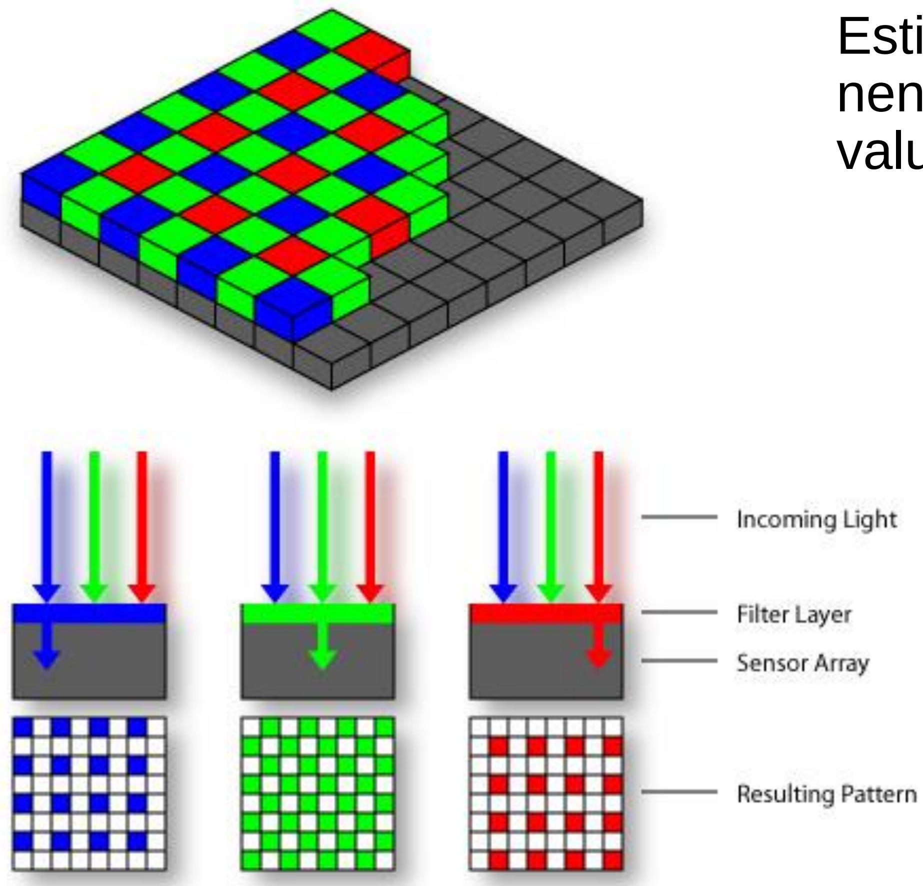
- As *f* gets larger, image becomes more *telescopic*
  - Smaller part of the world projects onto the finite image plane



from R. Duraiswami

Recap: Color Sensing in Digital Cameras

Bayer grid

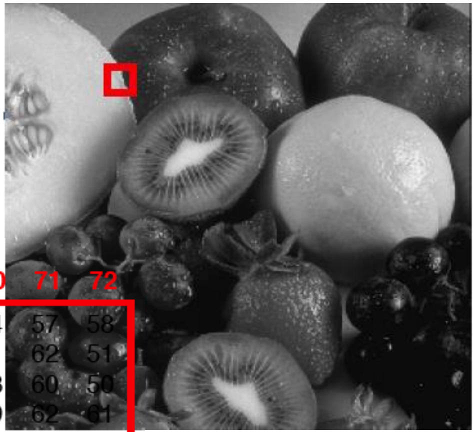


Estimate missing components from neighboring values (demosaicing)

Source: Steve Seitz

Recap: Images as Arrays of Numbers

- Problem of Computer Vision
  - How can we recognize fruits from an array of (gray-scale) numbers?
  - How can we perceive depth from an array of (gray-scale) numbers?
  - …



y = \ x =	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
41	210	209	204	202	197	247	143	71	64	80	84	54	54	57	58
42	206	196	203	197	195	210	207	56	63	58	53	53	61	62	51
43	201	207	192	201	198	213	156	69	65	57	55	52	53	60	50
44	216	206	211	193	202	207	208	57	69	60	55	77	49	62	61
45	221	206	211	194	196	197	220	56	63	60	55	46	97	58	106
46	209	214	224	199	194	193	204	173	64	60	59	51	62	56	48
47	204	212	213	208	191	190	191	214	60	62	66	76	51	49	55
48	214	215	215	207	208	180	172	188	69	72	55	49	56	52	56
49	209	205	214	205	204	196	187	196	86	62	66	87	57	60	48
50	208	209	205	203	202	186	174	185	149	71	63	55	55	45	56
51	207	210	211	199	217	194	183	177	209	90	62	64	52	93	52
52	208	205	209	209	197	194	183	187	187	239	58	68	61	51	56
53	204	206	203	209	195	203	188	185	183	221	75	61	58	60	60
54	200	203	199	236	188	197	183	190	183	196	122	63	58	64	66
55	205	210	202	203	199	197	196	181	173	186	105	62	57	64	63

  - *What does it mean to see? How can we make a computer do it?*

Slide credit: Michael Black

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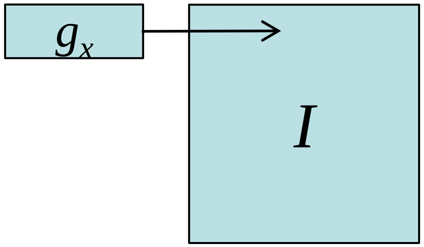
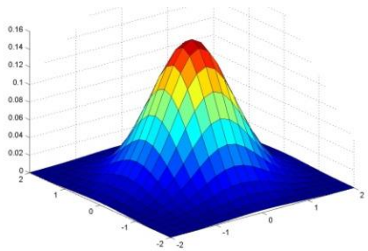
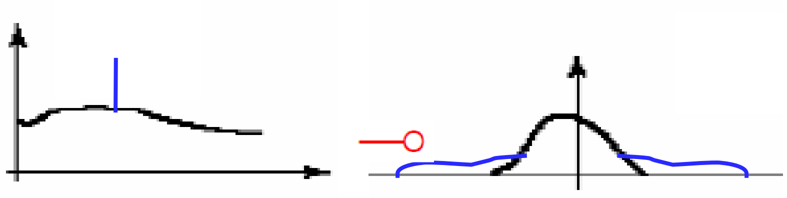


Linear Filters



Gaussian Smoothing



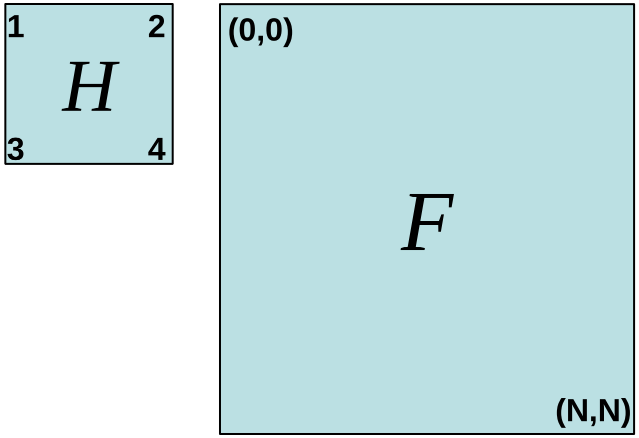
What does it mean to filter an image?

Recap: Correlation Filtering

$$G[i,j] = \sum_{u=-k}^{k} \sum_{v=-k}^{k} H[u,v] F[i+u, j+v]$$

- This is called cross-correlation, denoted $G = H \otimes F$

- Filtering an image
  - Replace each pixel by a weighted combination of its neighbors.
  - The filter "kernel" or "mask" is the prescription for the weights in the linear combination.



Slide credit: Kristen Grauman

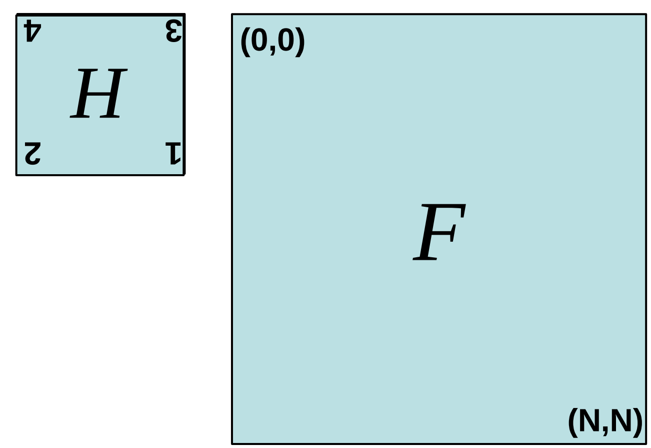
Recap: Convolution

- Convolution:
  - Flip the filter in both dimensions (bottom to top, right to left)
  - Then apply cross-correlation

$$G[i,j] = \sum_{u=-k}^{k} \sum_{v=-k}^{k} H[u,v]F[i-u, j-v]$$

$$G = H \star F$$

Notation for convolution operator



Slide credit: Kristen Grauman

Recap: Correlation vs. Convolution

- Correlation

$$G[i,j] = \sum_{u=-k}^{k} \sum_{v=-k}^{k} H[u,v]F[i+u,j+v]$$

$$G = H \otimes F$$

Note the difference!

- Convolution

$$G[i,j] = \sum_{u=-k}^{k} \sum_{v=-k}^{k} H[u,v]F[i-u,j-v]$$

$$G = H \star F$$

- Note
  - If $H[-u,-v] = H[u,v]$, then correlation $\equiv$ convolution.

Slide credit: Kristen Grauman

Recap: Properties of Convolution

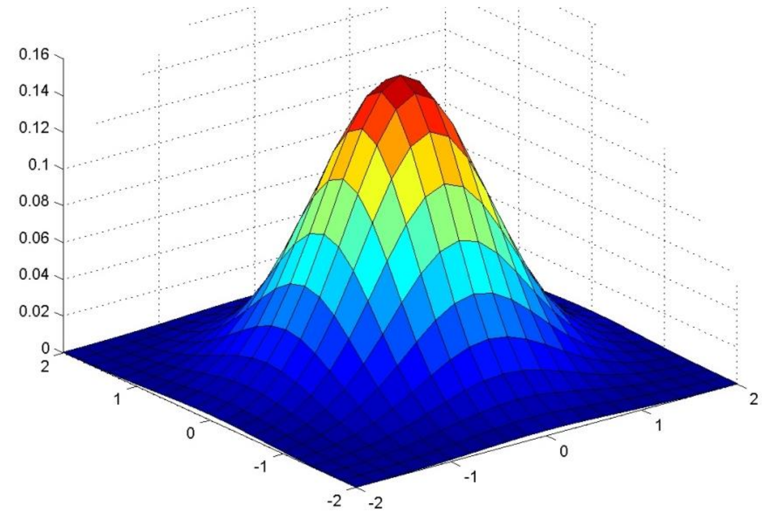
- Linear & shift invariant
  - Superposition: $h \star (f_1 + f_2) = (h \star f_1) + (h \star f_2)$
  - Scaling: $h \star (kf) = k(h \star f)$
- Commutative: $f \star g = g \star f$
- Associative: $(f \star g) \star h = f \star (g \star h)$
  - Often apply several filters in sequence: $(((a \star b_1) \star b_2) \star b_3)$
  - This is equivalent to applying one filter: $a \star (b_1 \star b_2 \star b_3)$
- Identity: $f \star e = f$
  - for unit impulse $e = [\ldots, 0, 0, 1, 0, 0, \ldots]$.
- Differentiation: $\frac{\partial}{\partial x}(f \star g) = \frac{\partial f}{\partial x} \star g$

Slide credit: Kristen Grauman

Recap: Gaussian Smoothing

- Gaussian kernel

$$G_\sigma = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$



- Rotationally symmetric
- Weights nearby pixels more than distant ones
  - This makes sense as 'probabilistic' inference about the signal
- A Gaussian gives a good model of a fuzzy blob

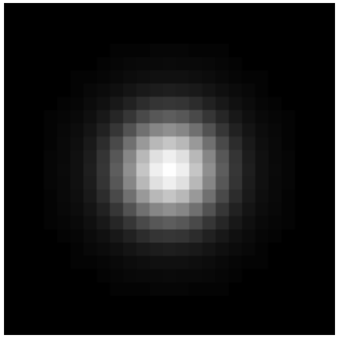
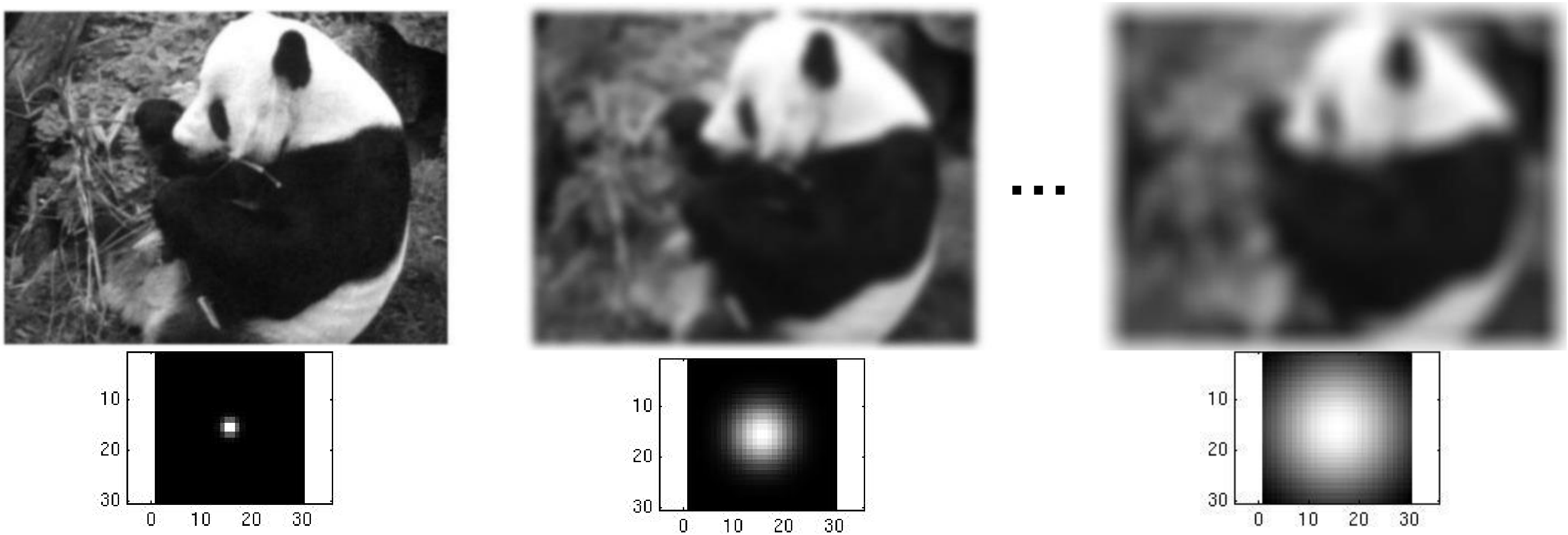


Image Source: Forsyth & Ponce

Recap: Smoothing with a Gaussian

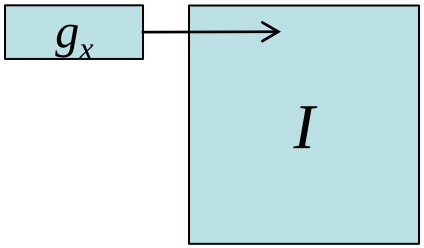
- Parameter σ is the “scale” / “width” / “spread” of the Gaussian kernel and controls the amount of smoothing.



Slide credit: Kristen Grauman

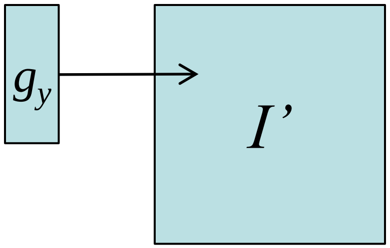
Recap: Efficient Implementation

- Both, the BOX filter and the Gaussian filter are separable:
  - First convolve each row with a 1D filter



$$g(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp(-x^2/(2\sigma^2))$$

  - Then convolve each column with a 1D filter



$$g(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp(-y^2/(2\sigma^2))$$

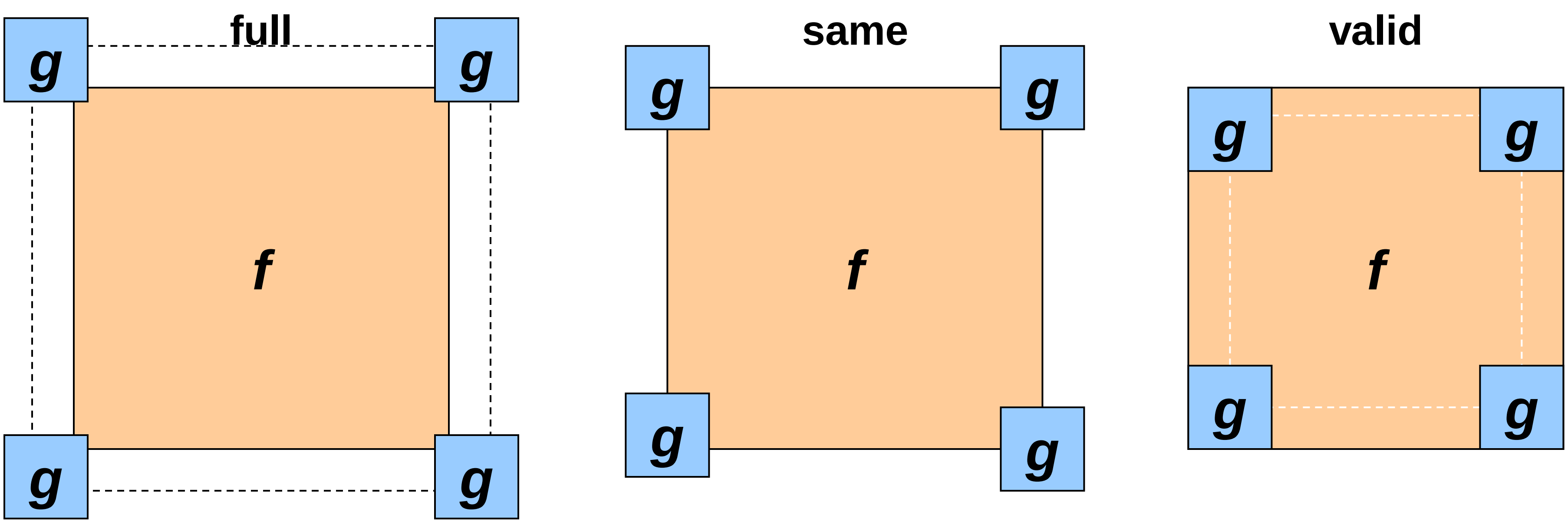
- Remember:
  - Convolution is linear – associative and commutative

$$g_x \star g_y \star I = g_x \star (g_y \star I) = (g_x \star g_y) \star I$$

Slide credit: Bernt Schiele

Filtering: Boundary Issues

- What is the size of the output?
- MATLAB: `filter2(g,f,shape)`
  - *shape* = 'full': output size is sum of sizes of f and g
  - *shape* = 'same': output size is same as f
  - *shape* = 'valid': output size is difference of sizes of f and g

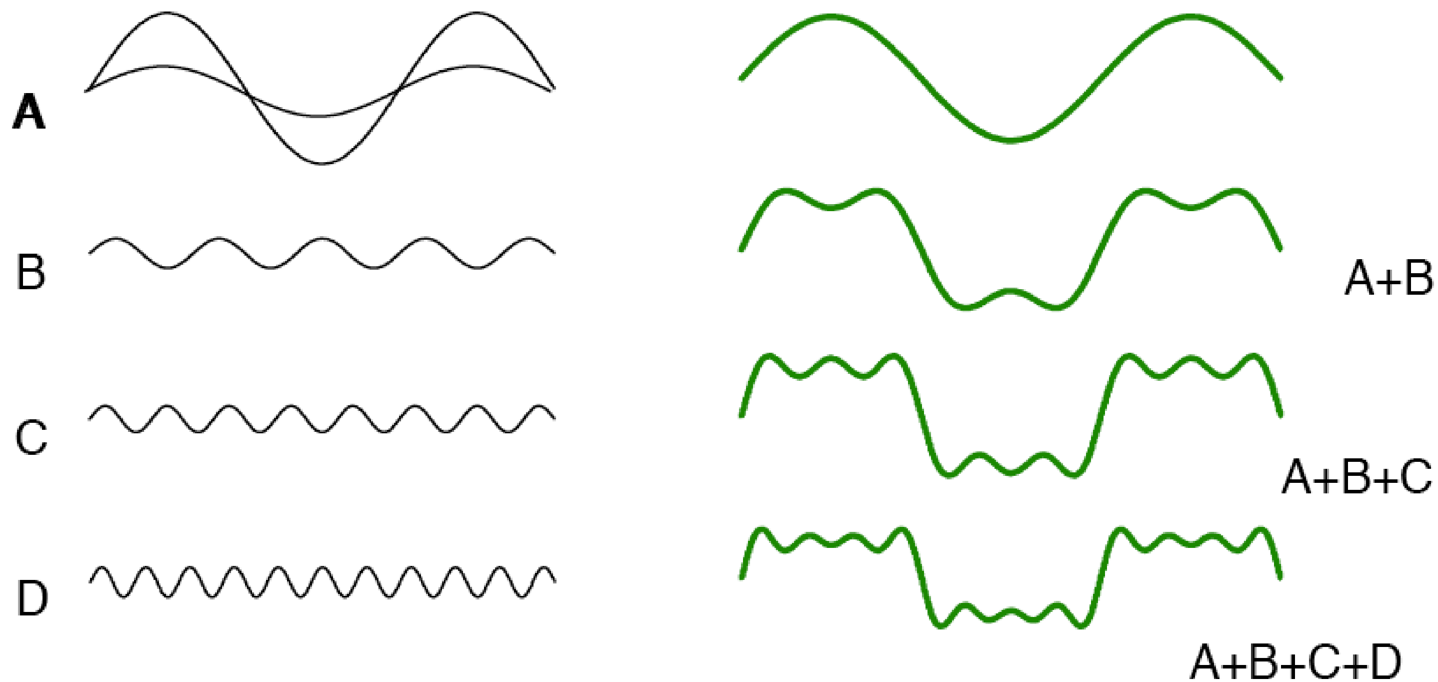


Slide credit: Svetlana Lazebnik

Why Does This Work?

- Let's take a small excursion into the Fourier domain to talk about spatial frequencies…

=



$3\cos(x)$

$+\ 1\cos(3x)$

$+\ 0.8\cos(5x)$

$+\ 0.4\cos(7x)$

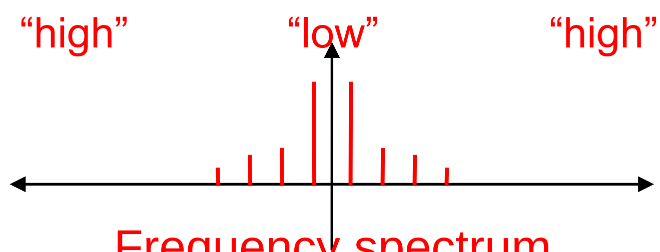
$+\ \ldots$

Source: Michal Irani

The Fourier Transform in Cartoons

- A small excursion into the Fourier transform to talk about spatial frequencies…

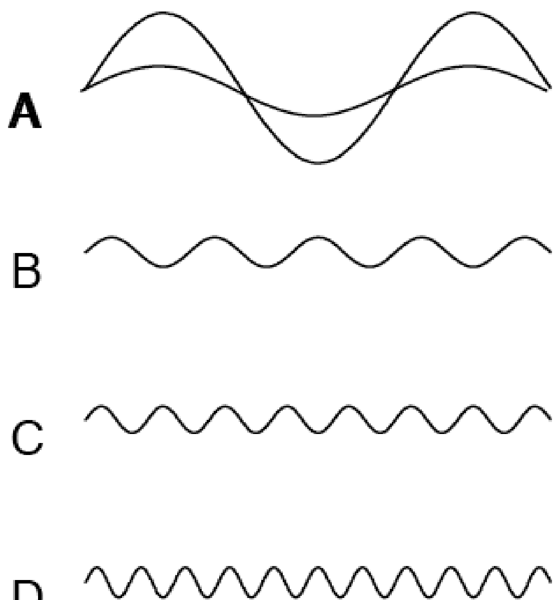
Fourier transform operator

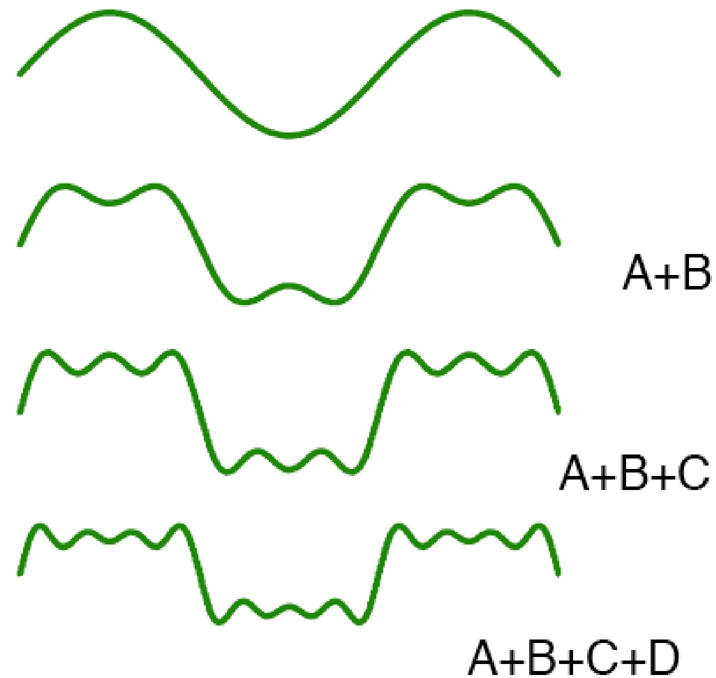


Frequency spectrum

$$3\cos(x) + 1\cos(3x) + 0.8\cos(5x) + 0.4\cos(7x) + \cdots$$

Frequency coefficients

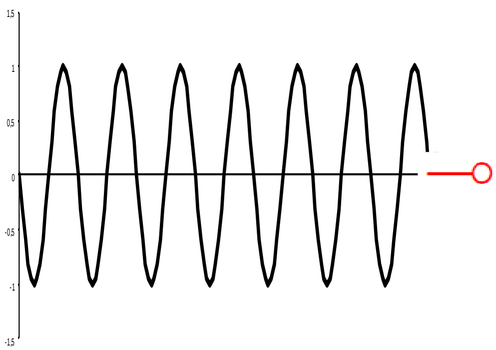




Source: Michal Irani

Fourier Transforms of Important Functions

- Sine and cosine transform to…



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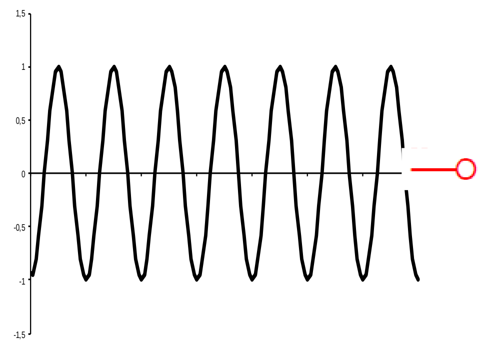

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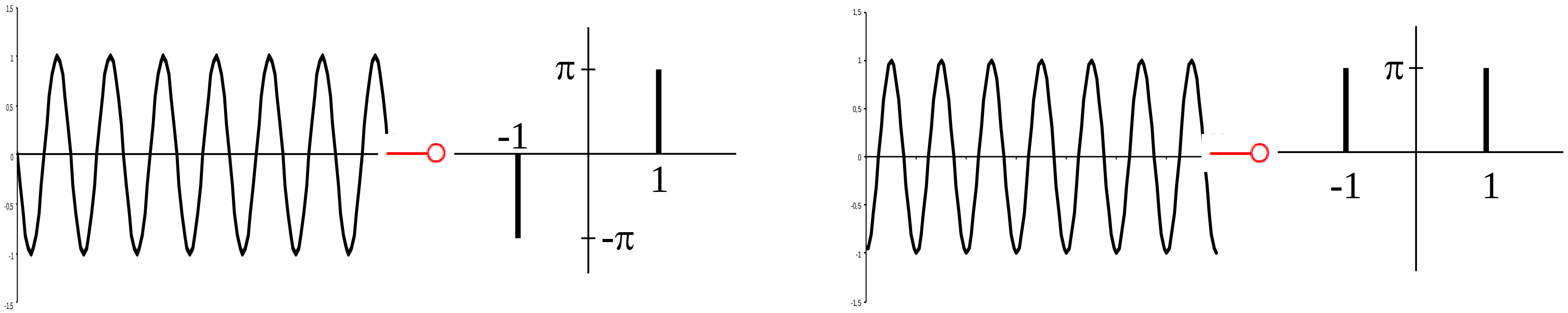
Image Source: S. Chenney

Fourier Transforms of Important Functions

- Sine and cosine transform to "frequency spikes"



- A Gaussian transforms to…

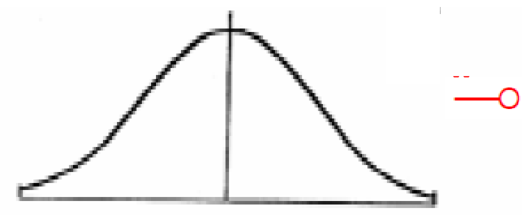


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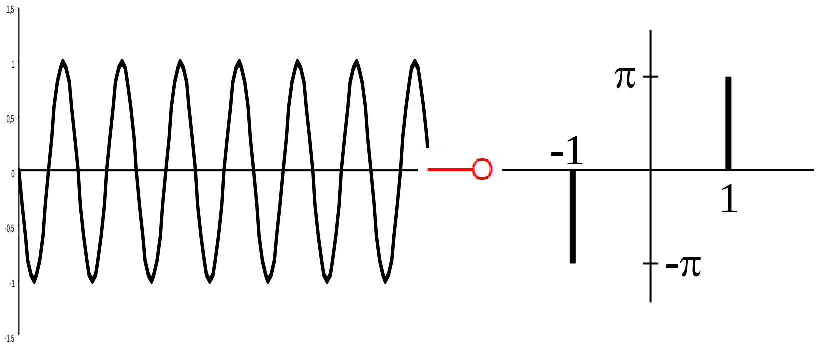
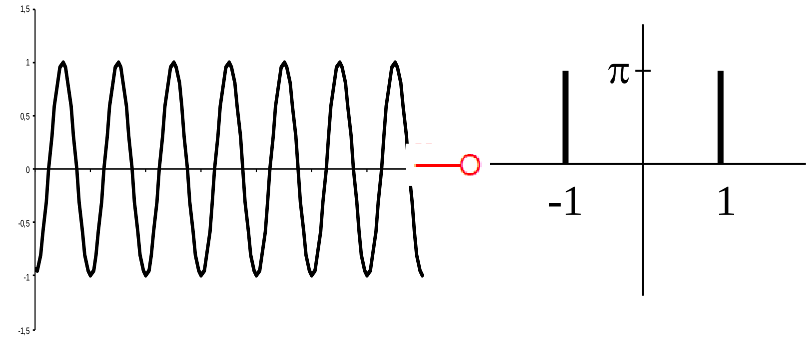
Image Source: S. Chenney

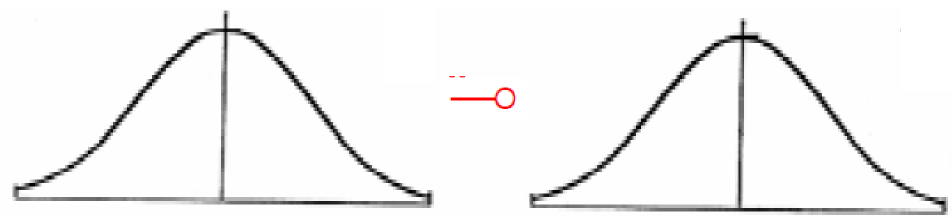
Fourier Transforms of Important Functions

- Sine and cosine transform to "frequency spikes"





- A Gaussian transforms to a Gaussian



- A box filter transforms to…

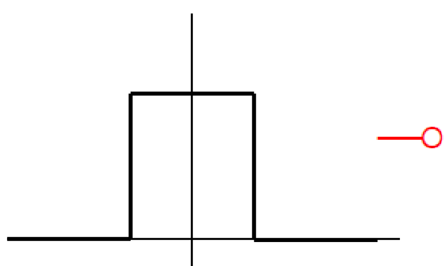


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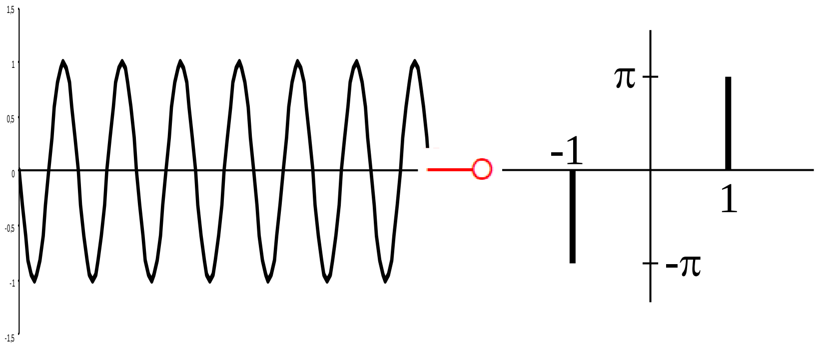
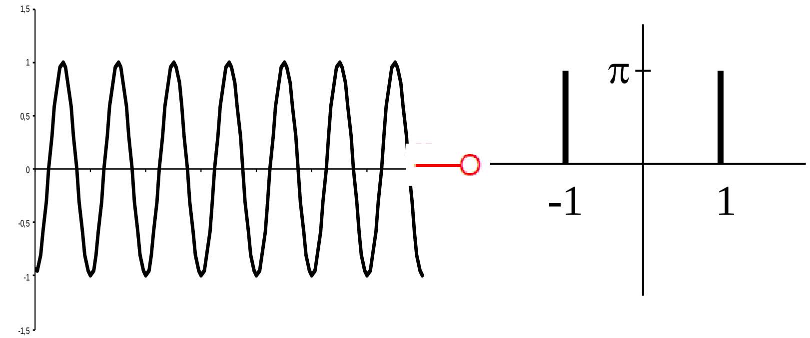
Image Source: S. Chenney

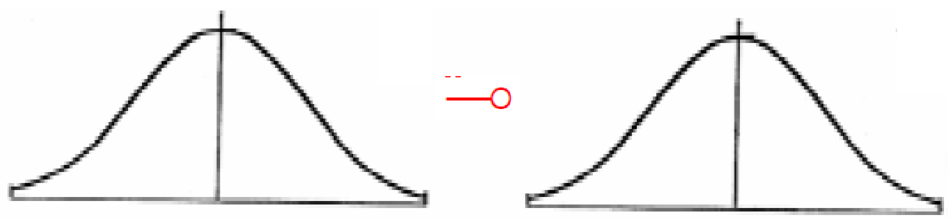
Fourier Transforms of Important Functions

- Sine and cosine transform to “frequency spikes”



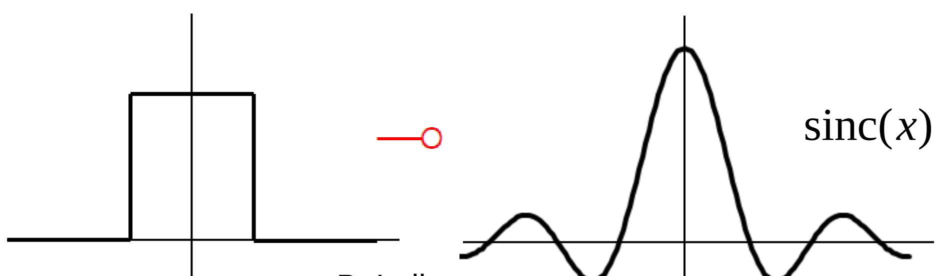


- A Gaussian transforms to a Gaussian



All of this is symmetric!

- A box filter transforms to a sinc

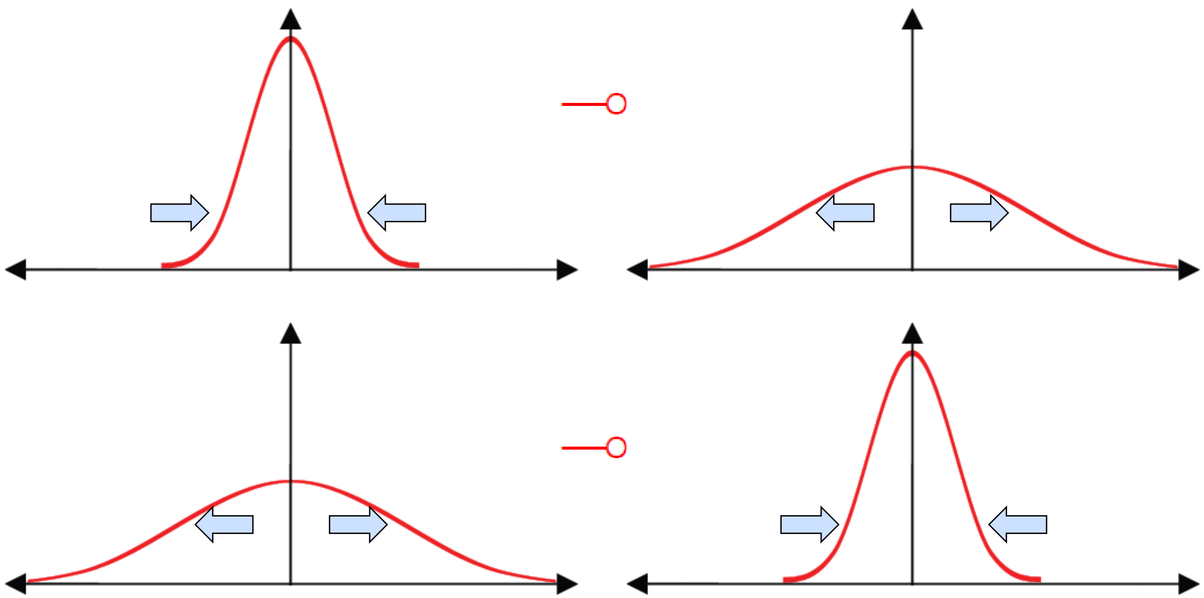


$$\operatorname{sinc}(x) = \frac{\sin x}{x}$$

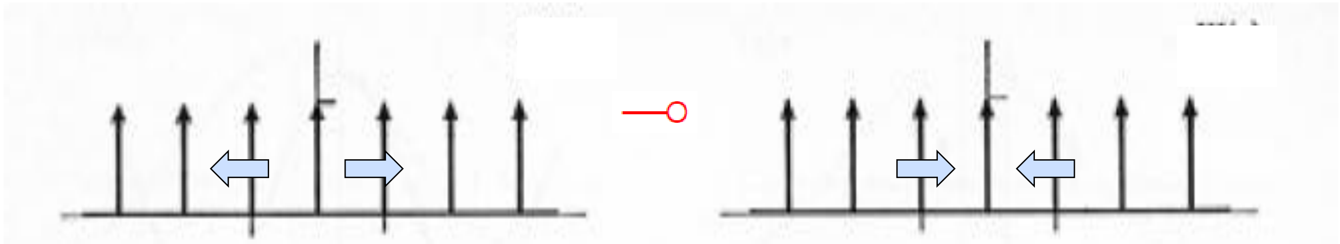
Image Source: S. Chenney

Duality

- The better a function is localized in one domain, the worse it is localized in the other.



- This is true for any function



Effect of Convolution

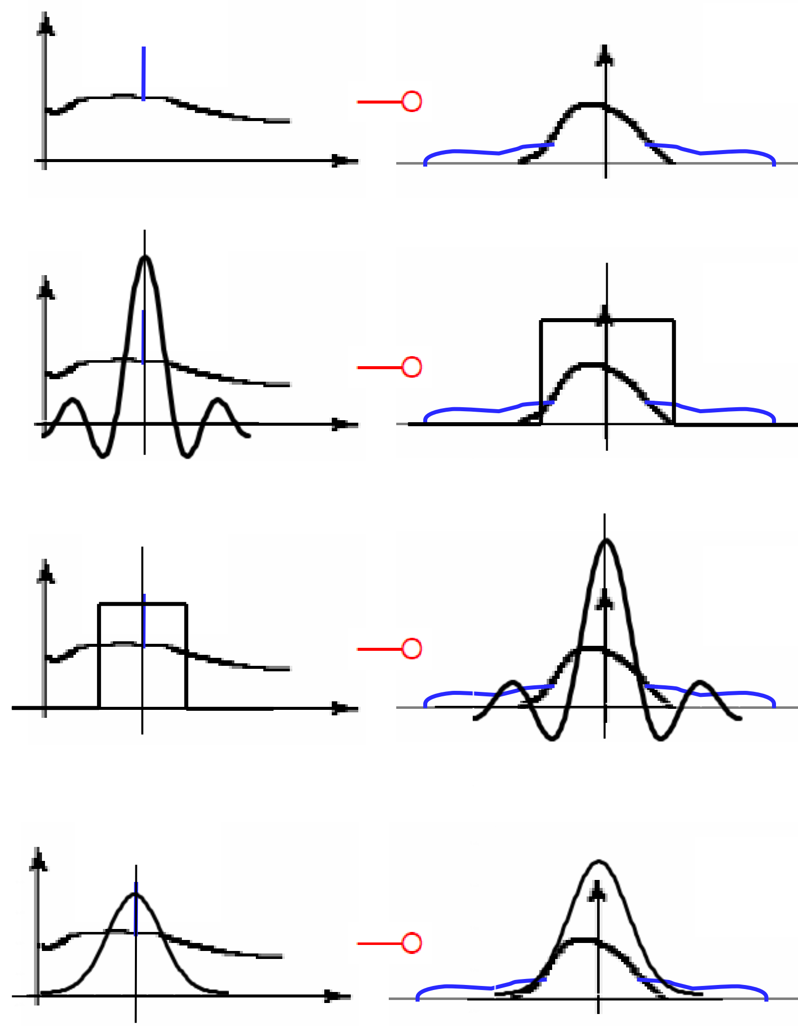
- Convolving two functions in the image domain corresponds to taking the product of their transformed versions in the frequency domain.

$$f \star g \multimap \mathcal{F} \cdot \mathcal{G}$$

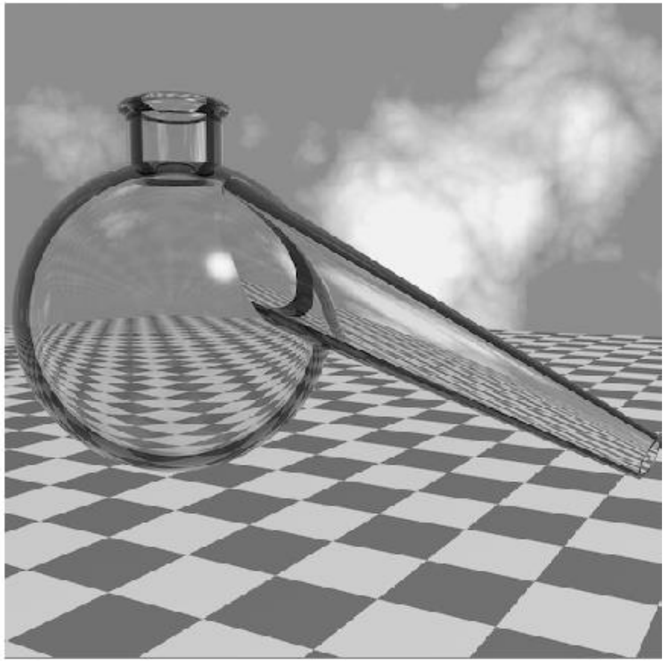
- This gives us a tool to manipulate image spectra.
  - A filter attenuates or enhances certain frequencies through this effect.

Recap: Effect of Filtering

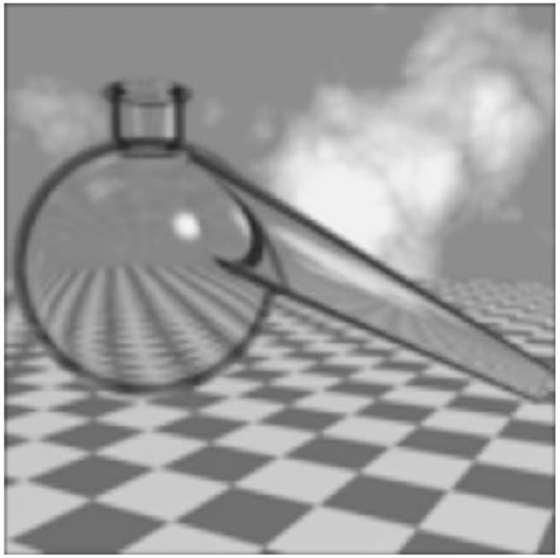
- Noise introduces high frequencies. To remove them, we want to apply a “low-pass” filter.
- The ideal filter shape in the frequency domain would be a box. But this transfers to a spatial sinc, which has infinite spatial support.
- A compact spatial box filter transfers to a frequency sinc, which creates artifacts.
- A Gaussian has compact support in both domains. This makes it a convenient choice for a low-pass filter.



Low-Pass vs. High-Pass



Original image



Low-pass filtered

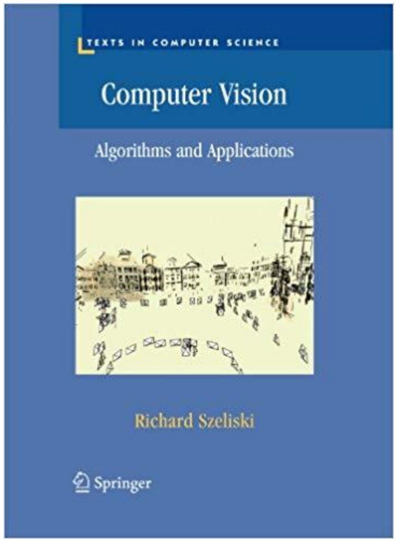
High-pass filtered

Image Source: S. Chenney

References and Further Reading

- Background information on linear filters and their connection with the Fourier transform can be found in Chapter 3 of the Szeliski book.

R. Szeliski
Computer Vision – Algorithms and Applications
Springer, 2010
(draft version available online)



Flipped Classroom Homework

- For the next lecture on Monday, please watch the following pre-recorded videos
  - Part 2.3: Non-linear Filters
  - Part 3.1: Multi-scale Representations
  - Part 3.2: Filters as Templates
  - Part 3.3: Image Gradients